ESP32 → CNC Shield V3

Az/El Satellite Rotator — Bench Wiring Reference

ESP32 Board	GPIO	CNC Shield	Signal
D25	**GPIO25**	**D2**	X STEP — Azimuth
D26	**GPIO26**	**D5**	X DIR — Azimuth
D27	**GPIO27**	**D3**	Y STEP — Elevation
D14	**GPIO14**	**D6**	Y DIR — Elevation
D13	**GPIO13**	**D8**	~EN (shared, active-LOW)
D32	**GPIO32**	**D9**	X Limit — Az home
D33	**GPIO33**	**D10**	Y Limit — El home
GND	**GND**	**GND**	**Common ground (MANDATORY)**

On the ESP32 the silkscreen D-numbers **are** the GPIO numbers (D25 = GPIO25). Wire each board pin straight to the matching shield label.

Firmware pin map

```
#define AZ_STEP 25      // D25 -> shield D2
#define AZ_DIR  26      // D26 -> shield D5
#define EL_STEP 27      // D27 -> shield D3
#define EL_DIR  14      // D14 -> shield D6
#define EN_PIN  13      // D13 -> shield D8   (LOW = enabled)
#define AZ_LIM  32      // D32 -> shield D9   (INPUT_PULLUP)
#define EL_LIM  33      // D33 -> shield D10  (INPUT_PULLUP)
```

Before you power up

- **Common ground first.** ESP32 GND must tie to a shield GND pin, or the steppers jitter or do nothing.
- **Keep motors disarmed at boot.** Add a pull-up on the EN line (or drive D13 HIGH immediately) so the drivers stay off until homing.
- **3.3 V logic is fine** for the DRV8825 — no level shifter needed.
- **Microstep jumpers** (M0/M1/M2 under each socket) set the stepping — all three ON = 1/32.
- **Set Vref per driver** for your NEMA 17 current *before* energising VMOT.
- **Power:** ESP32 over USB (share GND only); VMOT 12/24 V to the shield screw terminal with the bulk cap fitted.